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Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2021

Time: 03 Hours

Code: 101503 ***Subject: Geotechnical Engineering - I*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven): **[14]**

(a) Chemical weathering of rocks is characterized by: 1. change in chemical and mineralogical composition 2. disappearance of some minerals 3. formation of new secondary minerals. Correct answer is

(i) 1 only
(ii) 1 and 2 only
(iii) 2 and 3 only
(iv) 1, 2 and 3

(b) The bulk density of a soil mass below ground water table is equal to

(i) saturated density
(ii) submerged density
(iii) dry density
(iv) relative density

(c) The volume of a density bottle is usually

(i) 25 ml
(ii) 50 ml
(iii) 900 ml
(iv) 1000 ml

(d) Soil is in plastic state when the water content is between

(i) liquid limit and plastic limit
(ii) plastic limit and shrinkage limit
(iii) liquid limit and shrinkage limit
(iv) shrinkage limit and zero

(e) Wet sieve analysis is done for the soil passing

(i) 75 µm
(ii) 4.75 µm
(iii) 10 mm
(iv) 20 mm

(f) The maximum particle size for which Darcy's law is applicable is

(i) 75 µm
(ii) 0.2 mm
(iii) 0.5 mm
(iv) 1.0 mm

(g) A soil has a discharge velocity of 6×10^{-6}m/s and the porosity of 0.40. Its seepage velocity is

(i) 0.150×10^{-6}m/s
(ii) 1.5×10^{-6}m/s
(iii) 2.4×10^{-6}m/s
(iv) 6×10^{-6}m/s

(h) Quick sand is

(i) a type of sand
(ii) a condition in which a cohesionless soil losses its strength because of upward flow of water
(iii) a condition in which a cohesive soil losses its strength
(iv) None of the above

(i) The soil sample taken for IS light compaction test usually should pass through

(i) 4.75 mm IS sieve
(ii) 10 mm IS sieve
(iii) 20 mm IS sieve
(iv) 40 mm IS sieve

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(j) A point load of 700 kN is applied on the surface of a thick layer of saturated clay. Using Boussinesq's equation, the estimated vertical stress at a depth of 2 m and a radial distance of 1.0 m from the point of application of the load is

(i) 47.5 kPa
(ii) 47.7 kPa
(iii) 47.6 kPa
(iv) 47.8 kPa

Q.2 Solve both questions : **[14]**

(a) Explain the phenomena of formation and transportation of soils.

(b) Explain the method of determination of liquid limit of soil by Casagrande's apparatus. What are the limitations of the method?

Q.3 Solve both questions : **[14]**

(a) From fundamental, derive an expression for dry density in terms of bulk unit weight and water content.

(b) An earthen embankment under construction has a bulk unit weight of 17 kN/m³ and a moisture content of 12%. Compute the quantity of water in liters required to be added per cubic meter of earth to raise the moisture content to 16% at the same void ratio.

Q.4 Solve both questions : **[14]**

(a) Describe the Indian system of soil classification. When would you use dual symbols for soils?

(b) Two soils S_1 and S_2 are tested in the laboratory for the consistency limits. The data available is as follows:

	Soil S1	**Soil S2**
Plastic limit	21%	25%
Liquid limit	45%	65%
Flow index	8	10
Natural moisture content	40%	50%

(i) Which one is more plastic? (ii) Which one is better foundation material when remoulded? (iii) Which one has better strength as a function of water content? (iv) Which one has better strength at the plastic limit?

Q.5 Solve both questions : **[14]**

(a) What is flow net? Discuss the characteristics and uses of flow net.

(b) The following data were recorded in a constant head permeability test: Internal diameter of permeameter = 7.5 cm, Head loss over a sample length of 18 cm = 24.7 cm, Quantity of water collected 100 sec = 1000 ml, Porosity of soil sample was 45%. Calculate the coefficient of permeability and discharge velocity.

Q.6 Solve both questions : **[14]**

(a) Explain the mechanism of piping in hydraulic structures. What methods are used to increase the factor of safety against piping?

(b) A silt deposit consists of two layers. The top layer is 2.5 m thick (Unit weight = 18 kN/m³) and the bottom layer is 6.0 m thick (Unit weight = 19 kN/m³). The water table is at a depth of 4.0 m from the top. Draw the diagram showing the variation of total stress, neutral stress and effective stress.

Q.7 Solve both questions : **[14]**

(a) How the compaction improve the engineering properties of soil?

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(b) The following data is obtained in a compaction test: Bulk density (gm/cc): 1.97, 2.11, 2.42, 2.32, 2.30, 2.41. Moisture content (%): 8.81, 11.25, 14.05, 17.40, 19.25. Determine optimum moisture content and maximum dry density. Draw zero-air-void line. Assume G=2.68.

Q.8 Solve both questions : **[14]**

(a) Determine the vertical stress under a uniform circular load of intensity q and radius a using Westergaard's equation.

(b) The wall of a building foundation of width 4 m and length 30 m carries a uniformly distributed load 300 kN/m². Compute the vertical stress along the centre of the foundation at a depth 4.0 m below the bottom of the foundation. Ignore the surcharge effect of the soil over the base level of the foundation.

Q.9 Write short notes on any four of the following: **[14]**

(a) Black cotton soils
(b) Silt and clay
(c) Plasticity chart
(d) Piping
(e) Field compaction control
(f) Importance of pressure bulb

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